

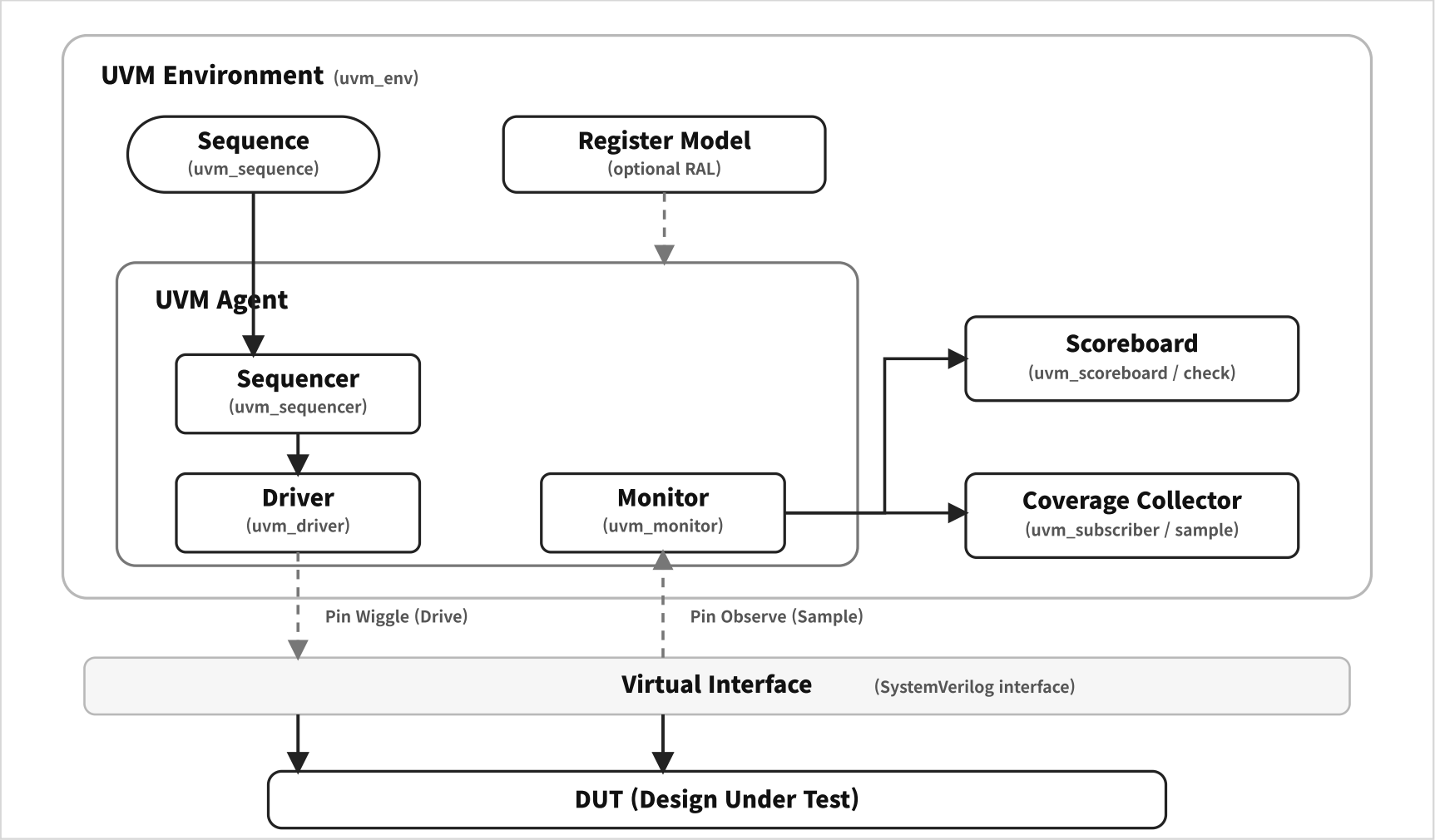
Adder, RAM, FIFO 검증 보고서

기본 RTL 블록을 대상으로 UVM testbench 구조, factory 기반 class 생성 흐름, sequence/scoreboard/coverage 구성, Verdi 결과 화면을 정리한다.

3 DUT Adder / RAM / FIFO	UVM test, env, agent, sequence	Checker scoreboard + coverage + SVA	Evidence Verdi waveform / result / coverage
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프로젝트 개요



UVM environment / agent / analysis / virtual interface 구조

검증 대상

DUT	검증 중심
Adder	조합 덧셈, carry bit
RAM	synchronous read/write
FIFO	ordering, full/empty/count

검증 방식

- sequence가 directed/random transaction 생성
- driver가 virtual interface로 DUT 구동
- monitor transaction을 scoreboard와 coverage로 전달
- RAM/FIFO는 protocol assertion 추가

공통 UVM Factory / Top Class 구조

factory 생성 흐름

```
env = *_env::type_id::create("env", this);
agt = *_agent::type_id::create("agt", this);
scb = *_scoreboard::type_id::create("scb", this);
cov = *_coverage::type_id::create("cov", this);

seq = *_sequence::type_id::create("seq");
seq.start(env.agt.sequencer);
```

top class 역할

Class	역할
test	env 생성, sequence 실행, objection 제어
env	agent, scoreboard, coverage 연결
agent	sequencer, driver, monitor 구성

***_test**
run_phase에서 sequence 시작

↓ **factory create**

***_env**
agent / scoreboard / coverage 생성

↓ **build_phase**

***_agent**
active agent

***_scoreboard**
expected 비교

***_coverage**
covergroup sample

↓ **agent 내부**

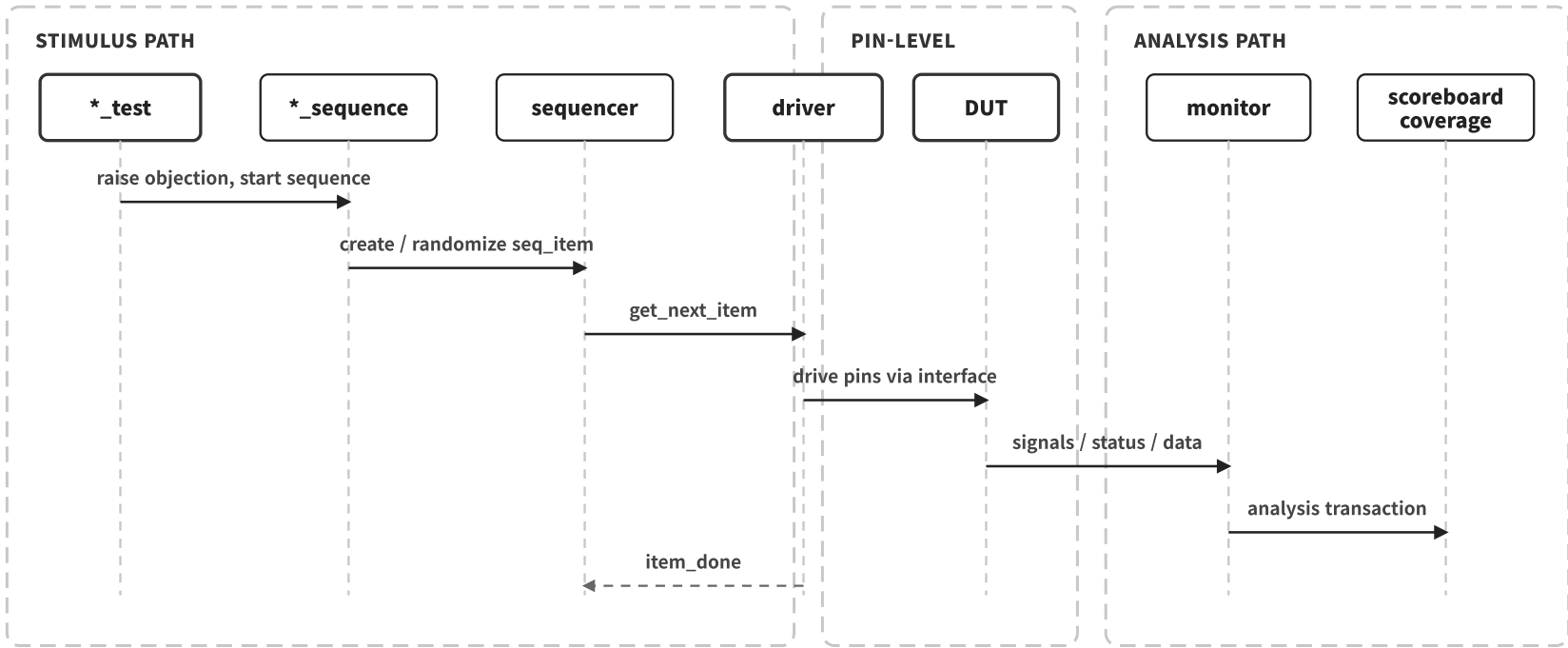
***_sequencer**
transaction 중재

***_driver**
DUT pin drive

***_monitor**
transaction 복원

Figure 2. Common UVM Transaction Sequence

Stimulus path, DUT boundary, and analysis path shown as lifelines instead of process boxes.



Adder, RAM, and FIFO keep this topology; only timing, fields, reference model, and coverage bins differ.

sequence → driver → DUT → monitor → scoreboard/coverage

Stimulus path

- test가 sequence 시작
- sequence가 seq_item 생성 및 randomize
- sequencer가 driver에 item 전달

Analysis path

- monitor가 DUT signal을 transaction으로 복원
- scoreboard가 reference model과 비교
- coverage가 command/data/status bin sample

Adder RTL

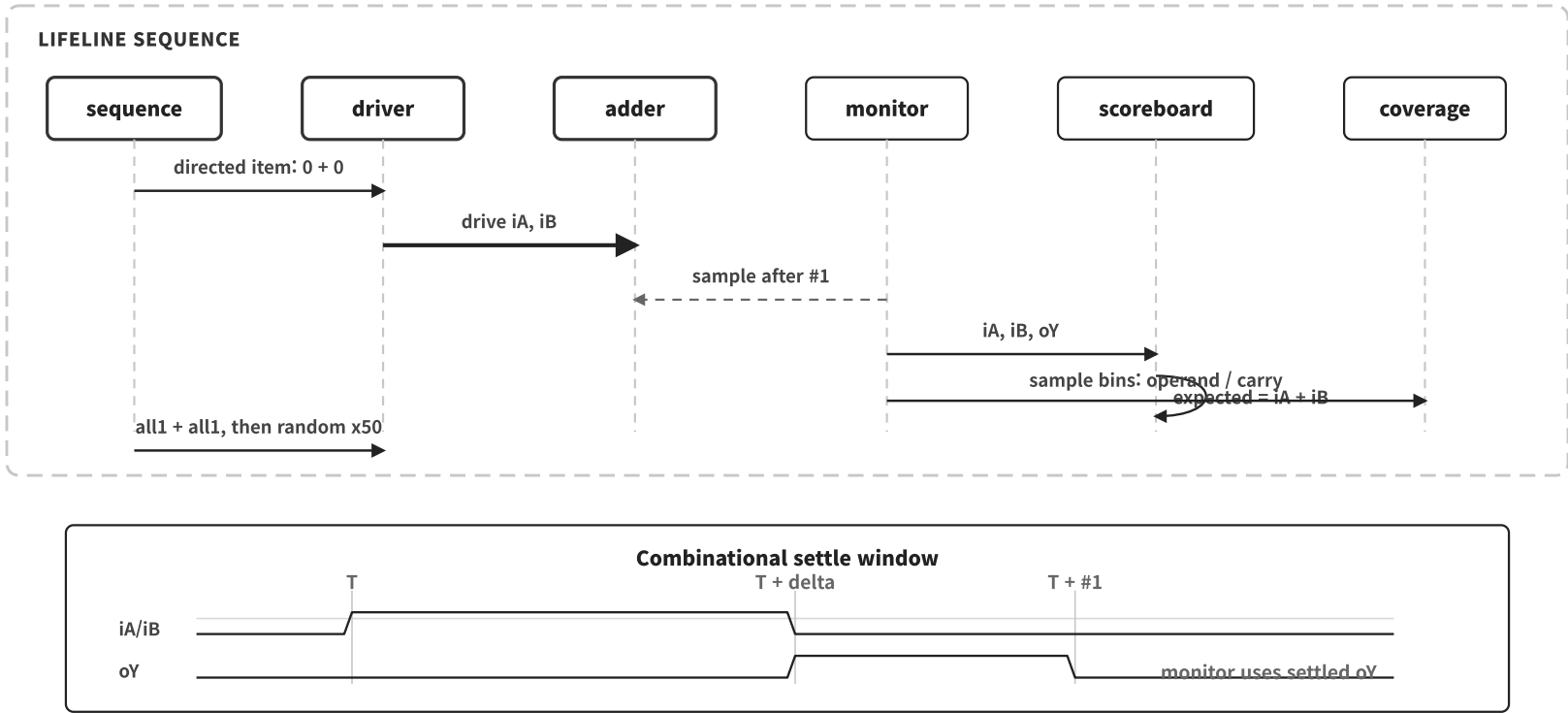
모듈 동작

Signal	설명
iA, iB	DATA_WIDTH bit 입력 operand
oY	DATA_WIDTH+1 bit 출력, MSB는 carry

```
module adder #(DATA_WIDTH = 32) (  
    input logic [DATA_WIDTH-1:0] iA,  
    input logic [DATA_WIDTH-1:0] iB,  
    output logic [DATA_WIDTH :0] oY  
);  
    assign oY = iA + iB;  
endmodule
```

Figure 3. Adder UVM Sequence and Combinational Timing

The monitor samples after #1 so the combinational output has settled before scoreboard comparison.

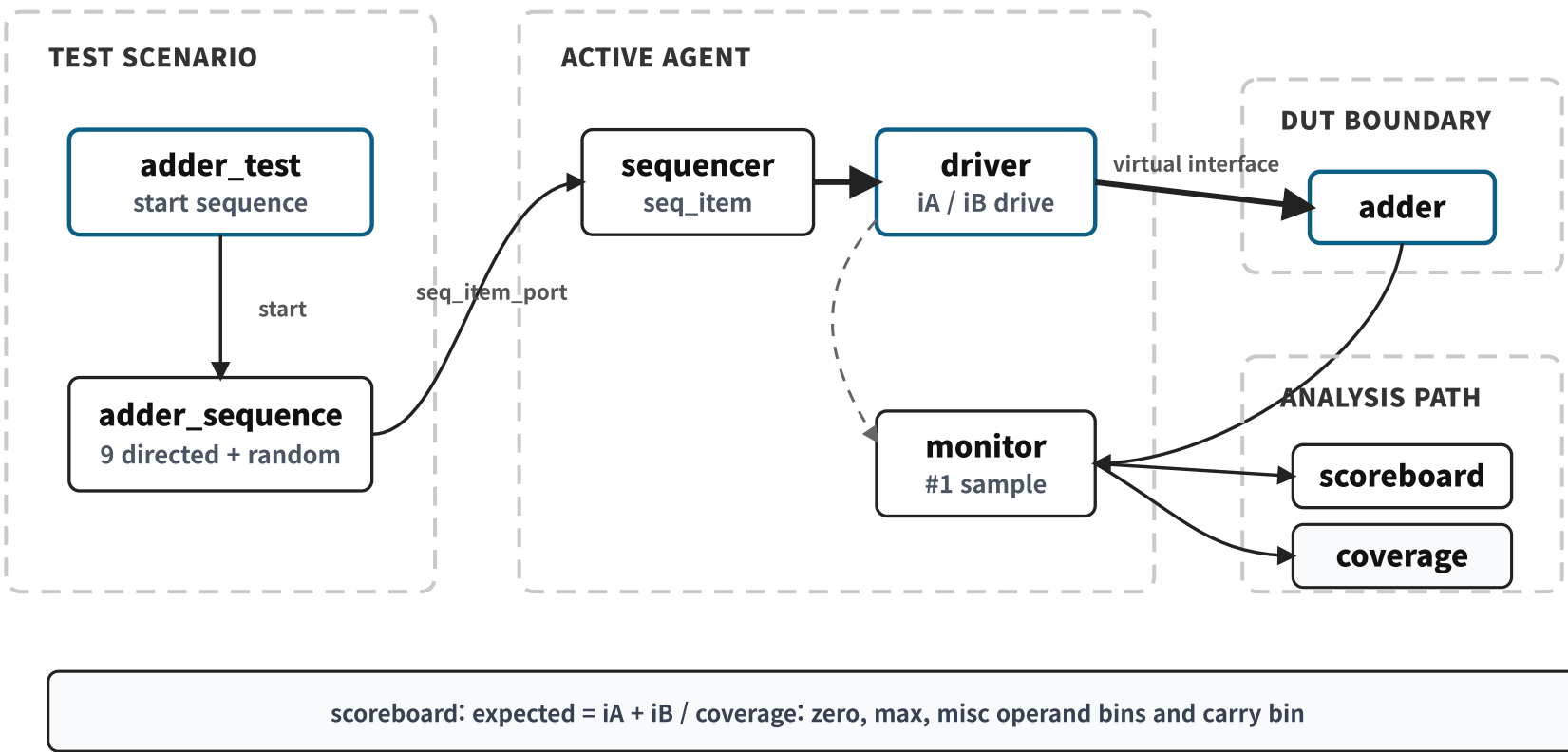


Adder combinational sampling timing

Adder UVM Class 구조

Adder UVM Class Map

combinational DUT: drive operands, sample after settle, compare iA+iB



Adder UVM class map

connect_phase

```
agtl.monitor.send.connect(scb.recv);
agtl.monitor.send.connect(cov.analysis_export);
driver.seq_item_port.connect(
    sequencer.seq_item_export
);
```

Adder에서 달라지는 점

- clocking block 없이 조합 settle 후 sample
- scoreboard reference model은 상태가 없는 즉시 계산
- coverage는 입력 corner와 carry 발생 여부 중심

Adder Sequence / Coverage

sequence 구성

구간	내용
directed	zero / misc / max 조합 9개
random	50개 transaction
핵심 case	max + max carry 발생

coverage bins

- cp_iA, cp_iB: zero / max / misc
- cp_carry: no_carry / carry
- cross_iA_iB: operand 조합

```
UVM_INFO adder_coverage.sv(45) @ 69000: uvm_test_top.env.cov [ADDER_COV] Functional coverage = 100.00%
UVM_INFO /tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_report_server.svh(904) @ 69000: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

** Report counts by severity
UVM_INFO :    73
UVM_WARNING :    0
UVM_ERROR :    0
UVM_FATAL :    0
** Report counts by id
[ADDER_COV]      1
[ADDER_SCB]     69
[RNTST]          1
[TEST_DONE]      1
[UVM/RELNOTES]   1

$finish called from file "/tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_root.svh", line 527.
$finish at simulation time                69000

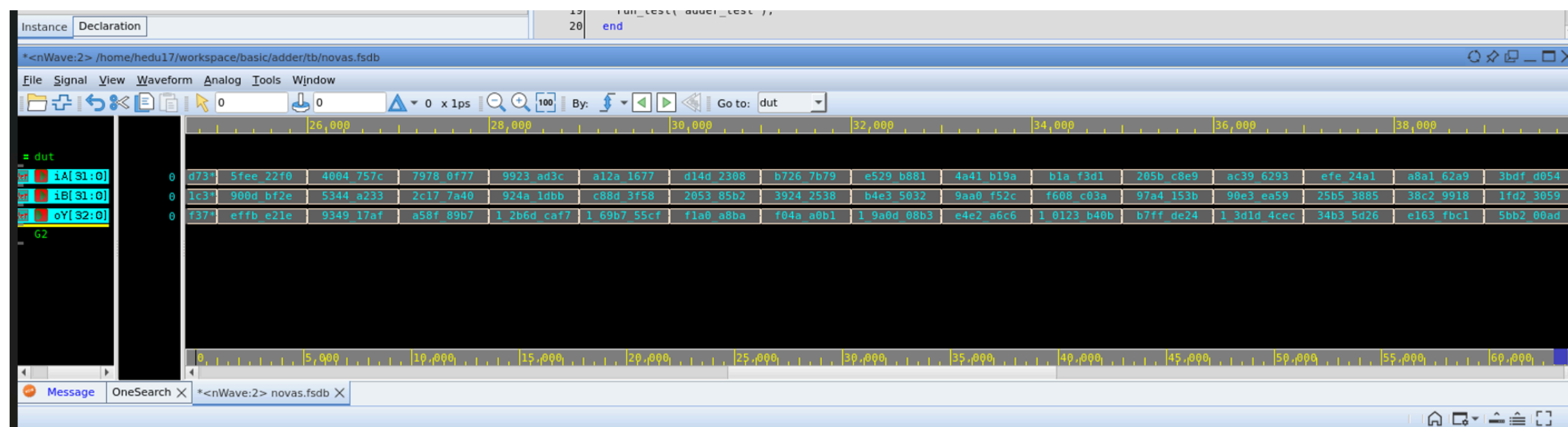
-----
VCS Coverage Metrics: during simulation line, cond, FSM, branch, tgl was monitored
-----

Coverage status: End of All Coverages ...

          V C S   S i m u l a t i o n   R e p o r t
Time: 69000 ps
CPU Time:      0.210 seconds;      Data structure size:  0.6Mb
Mon May 11 15:13:59 2026
```

Adder coverage / result evidence

Adder 결과



Adder Verdi waveform

Summary								
Hierarchy Modules Groups Asserts Statistics Tests								
Avg. Group Score:100.00% U+C:10 U:0 C:10 X:0								
Avg. Group Inst. Score:100.00% U+C:10 U:0 C:10 X:0								
Group	Score	Instances	U+C	U	C	X		
tb_adder_pkg::adder_cov...	100.00%	100.00%	10	0	10			
cp_carry	100.00%		2	0	2			
cp_iA	100.00%		2	0	2			
cp_iB	100.00%		2	0	2			
cross_iA_iB	100.00%		4	0	4			
tb_adder.me.obj.add...	100.00%		10	0	10			

UVM result summary

```
UVM_INFO adder_coverage.sv(45) @ 69000: uvm_test_top.env.cov [ADDER_COV] Functional coverage = 100.00%
UVM_INFO /tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_report_server.svh(904) @ 69000: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

** Report counts by severity
UVM_INFO : 73
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0
** Report counts by id
[ADDER_COV] 1
[ADDER_SCB] 69
[RNTST] 1
[TEST_DONE] 1
[UVM/RELNOTES] 1

$finish called from file "/tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_root.svh", line 527.
$finish at simulation time 69000

-----
VCS Coverage Metrics: during simulation line, cond, FSM, branch, tgl was monitored
-----

Coverage status: End of All Coverages ...

VCS Simulation Report
Time: 69000 ps
CPU Time: 0.210 seconds; Data structure size: 0.6Mb
Mon May 11 15:13:59 2026
```

Coverage result

RAM RTL

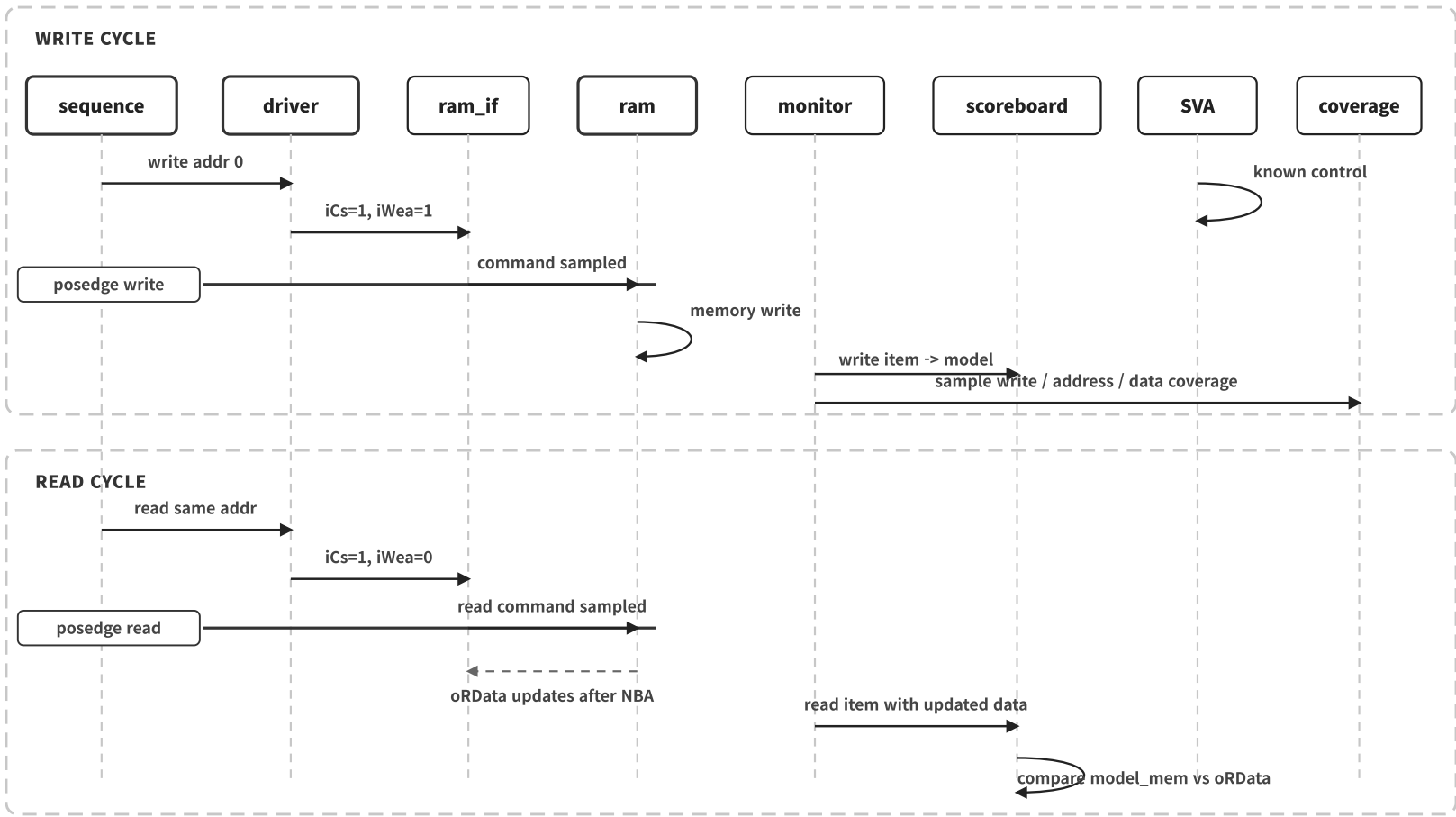
동작 조건

조건	동작
!iRstn	oRData와 memory 초기화
iCs && iWea	mem[iAddr] <= iWData
iCs && !iWea	oRData <= mem[iAddr]

```
always_ff @(posedge iClk or negedge iRstn) begin
  if (!iRstn) begin
    oRData <= '0;
    mem[idx] <= '0;
  end else if (iCs) begin
    if (iWea) mem[iAddr] <= iWData;
    else      oRData <= mem[iAddr];
  end
end
```

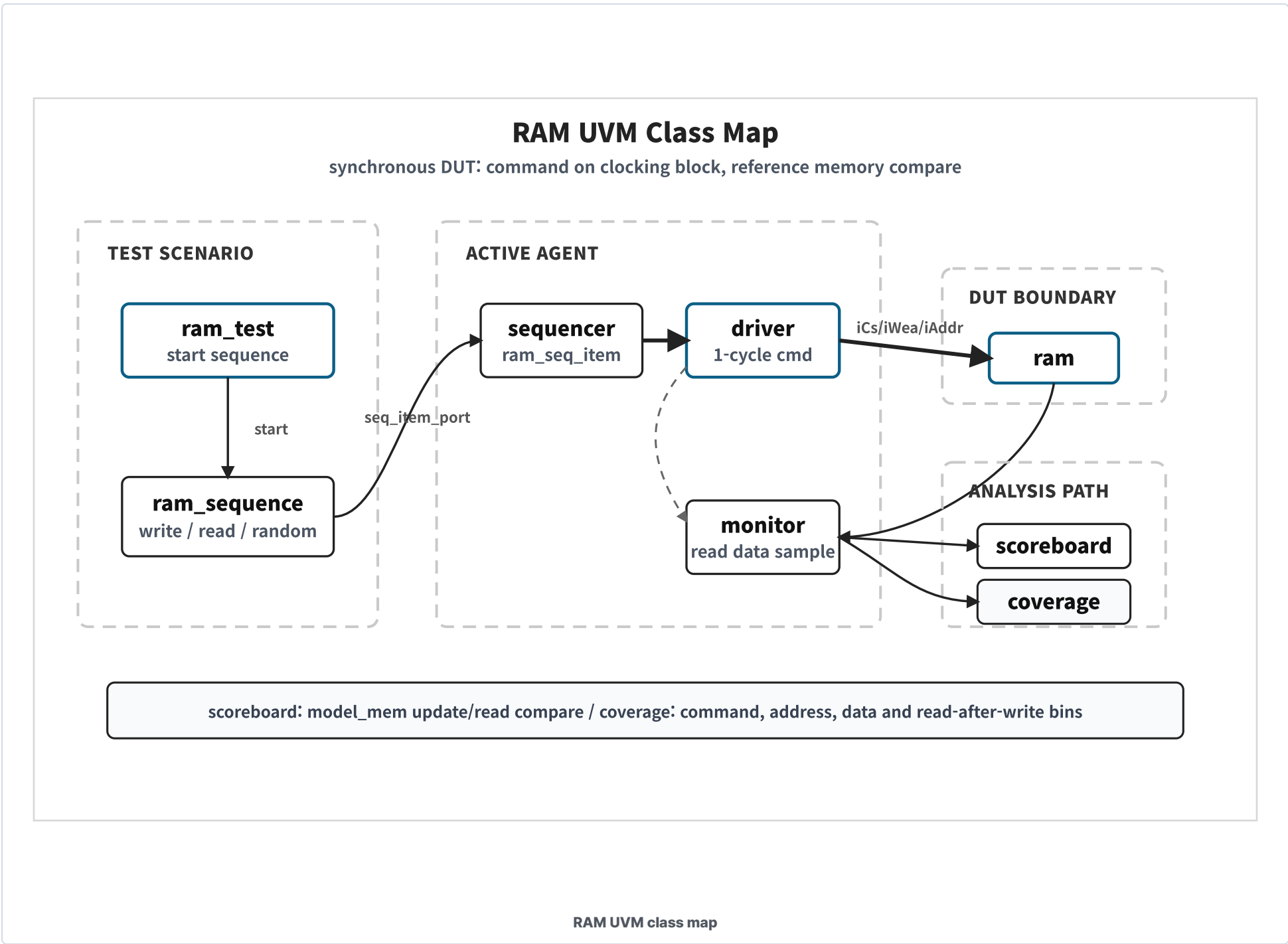
Figure 4. RAM UVM Sequence and Synchronous Timing

Write/read commands are sampled on the active clock edge; read data is checked after oRData settles.



RAM synchronous read/write timing

RAM UVM Class 구조



RAM에서 필요한 처리

- driver는 clocking block으로 command 유지
- monitor는 read command 이후 갱신된 oRData sample
- scoreboard는 reference memory 상태 유지

interface SVA

- selected transaction에서 X/Z control 방지
- write data known check
- reset 중 command idle check

RAM Sequence / Scoreboard / SVA

sequence 구성

구간	내용
directed	first / last / misc address write-read
sweep	0부터 DEPTH-1까지 write/read
random	100개 random access

scoreboard model

```
write: model_mem[addr] = iWData
read : expected = model_mem[addr]
      compare expected with oRData
```

coverage/SVA

- cp_cmd: read / write
- cp_addr: first / last / misc
- cross_cmd_addr + control/data assertion

```
Ready to proceed to the extract phase
UVM_INFO ram_coverage.sv(50) @ 2815000: uvm_test_top.env.cov [RAM_COV] Functional coverage = 100.00%
UVM_INFO /tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_report_server.svh(904) @ 2815000: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

** Report counts by severity
UVM_INFO : 134
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0
** Report counts by id
[RAM_COV] 1
[RAM_SCB] 130
[RNTST] 1
[TEST_DONE] 1
[UVM/RELNOTES] 1

$finish called from file "/tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_root.svh", line 527.
$finish at simulation time 2815000

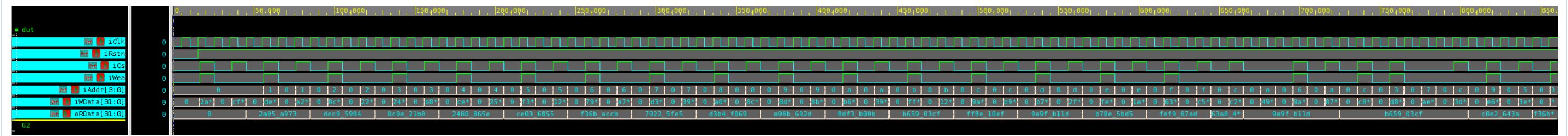
-----
VCS Coverage Metrics: during simulation line, cond, FSM, branch, tgl was monitored
-----

Coverage status: End of All Coverages ...

VCS Simulation Report
Time: 2815000 ps
CPU Time: 0.230 seconds; Data structure size: 0.6Mb
```

RAM coverage / result evidence

RAM 결과



RAM Verdi waveform

Group	Score	Instances	U+C	U	C	X
Cg tb_ram_pkg::ram_covera...	100.00%	100.00%	12	0	12	
CP cp_addr	100.00%		2	0	2	
CP cp_cmd	100.00%		2	0	2	
CP cp_rdata	100.00%		2	0	2	
CP cp_wdata	100.00%		2	0	2	
CR cross_cmd_addr	100.00%		4	0	4	
Cg tb_ram.me.obj.ram_cg	100.00%		12	0	12	

UVM result summary

```
UVM_INFO ram_coverage.sv(50) @ 2815000: uvm_test_top.env.cov [RAM_COV] Functional coverage = 100.00%
UVM_INFO /tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_report_server.svh(904) @ 2815000: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

** Report counts by severity
UVM_INFO : 134
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0
** Report counts by id
[RAM_COV] 1
[RAM_SCB] 130
[RNTST] 1
[TEST_DONE] 1
[UVM/RELNOTES] 1

$finish called from file "/tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_root.svh", line 527.
$finish at simulation time 2815000

-----
VCS Coverage Metrics: during simulation line, cond, FSM, branch, tgl was monitored
-----

Coverage status: End of All Coverages ...

VCS Simulation Report
Time: 2815000 ps
CPU Time: 0.230 seconds; Data structure size: 0.6Mb
```

Coverage result

FIFO RTL

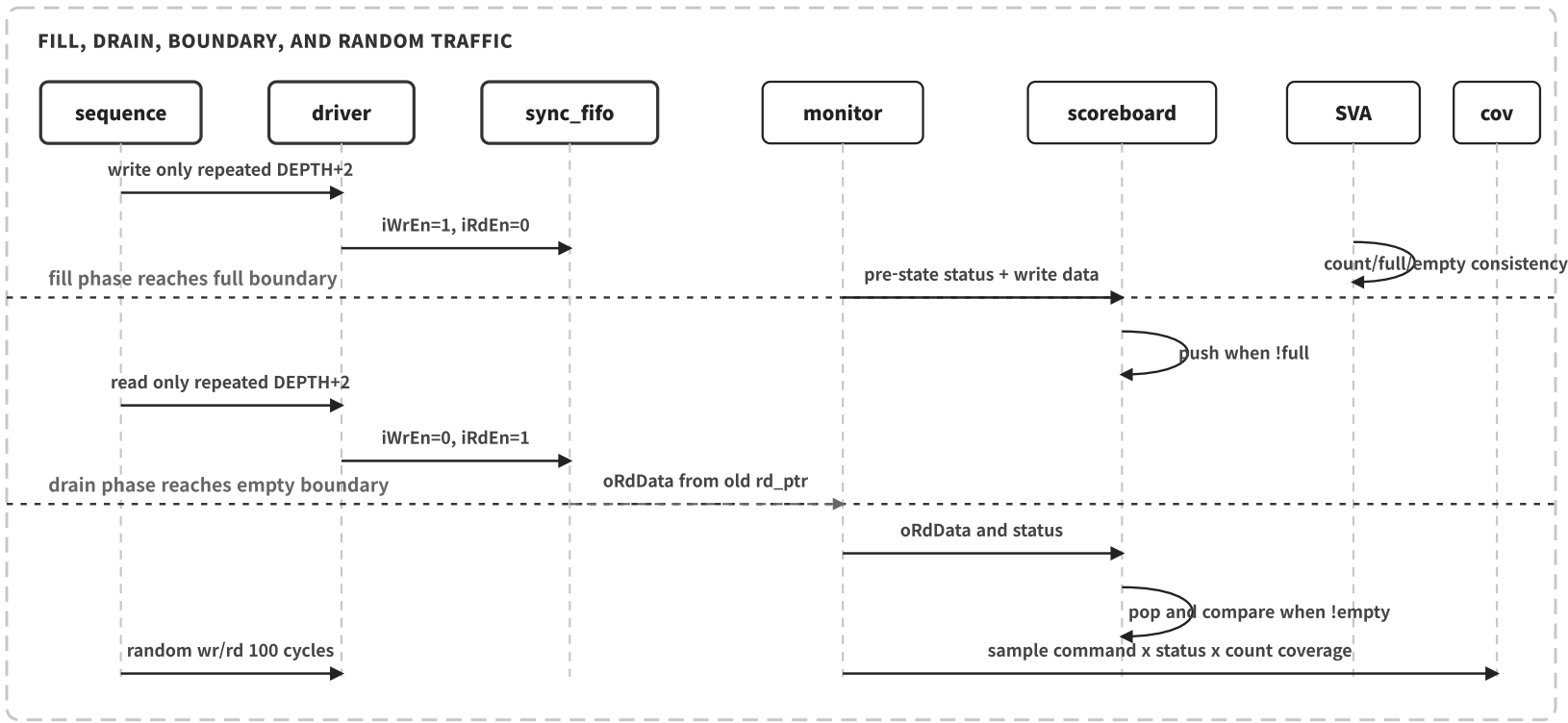
상태 신호

Signal	기준
oFull	oCount == DEPTH
oEmpty	oCount == 0
oRdData	mem[rd_ptr]

```
unique case ({iWrEn && !oFull,
             iRdEn && !oEmpty})
  2'b10: write, count++
  2'b01: read,  count--
  2'b11: write and read, count hold
default: state hold
endcase
```

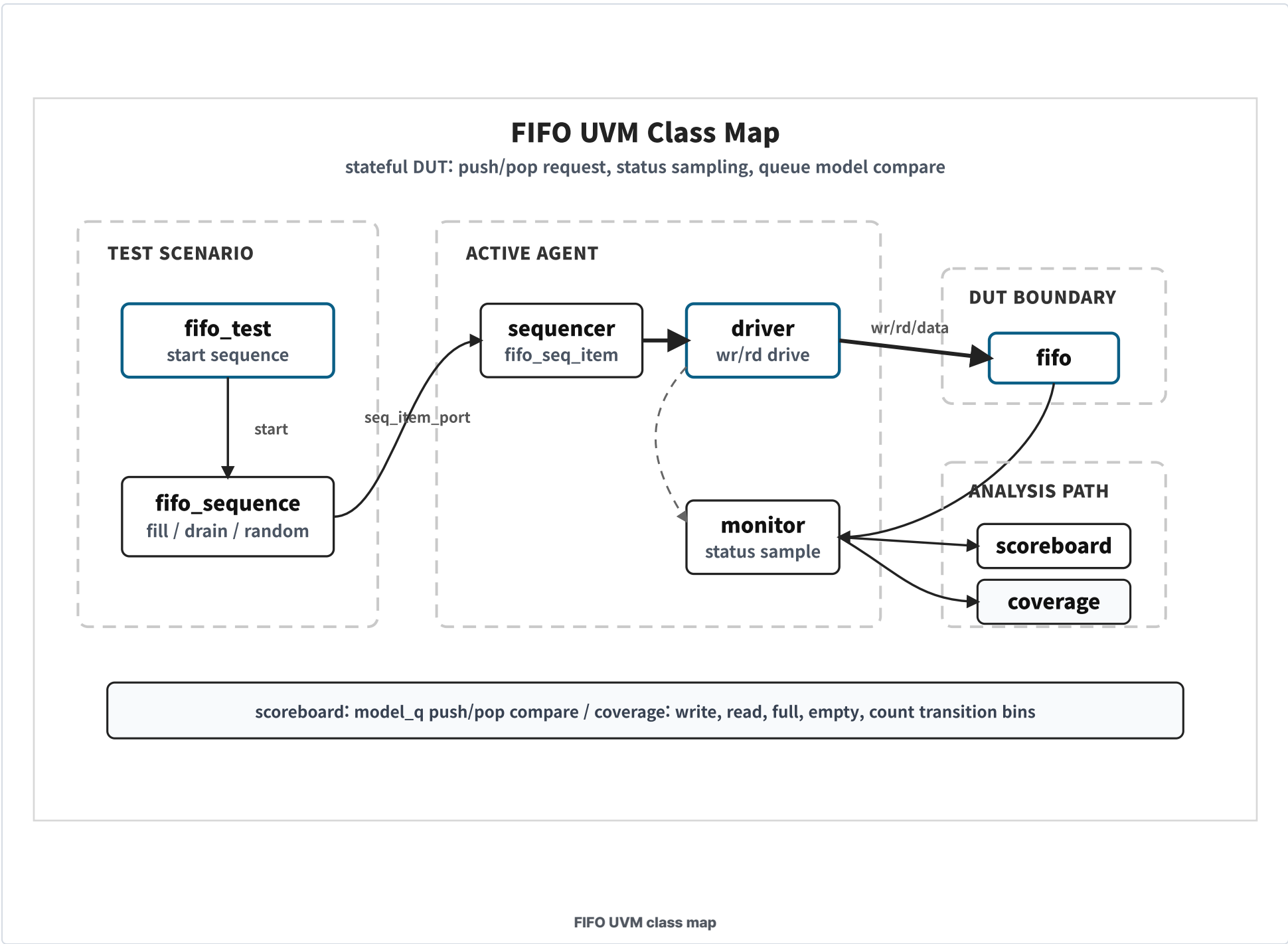
Figure 5. FIFO UVM Sequence and Queue Reference Model

The scoreboard decides wr_fire and rd_fire from pre-state full/empty status sampled by the monitor.



Blocked requests are modeled explicitly: full write is ignored, empty read is ignored, simultaneous valid write/read keeps count stable.

FIFO UVM Class 구조



FIFO에서 필요한 처리

- read data는 edge 직전 old rd_ptr 기준으로 확인
- scoreboard는 full/empty를 이용해 fire 여부 계산
- count/full/empty consistency assertion 사용

blocked request

- full 상태 write는 push하지 않음
- empty 상태 read는 pop하지 않음
- valid write/read 동시에는 count 유지

FIFO Sequence / Scoreboard / Coverage

sequence 구성

구간	내용
fill	write only DEPTH+2
drain	read only DEPTH+2
mixed	simultaneous write/read + random 100

scoreboard model

```
wr_fire = iWrEn && !oFull
rd_fire = iRdEn && !oEmpty

if (rd_fire) compare(model_q.pop_front(), oRdData)
if (wr_fire) model_q.push_back(iWrData)
```

coverage bins

- cp_cmd: idle/write/read/wr_rd
- cp_count: zero/mid/full
- cross_cmd_status

```
UVM_INFO fifo_coverage.sv(50) @ 2515000: uvm_test_top.env.cov [FIFO_COV] Functional coverage = 100.00%
UVM_INFO /tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_report_server.svh(904) @ 2515000: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

** Report counts by severity
UVM_INFO : 123
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0
** Report counts by id
[FIFO_COV] 1
[FIFO_SCB] 119
[RNTST] 1
[TEST_DONE] 1
[UVM/RELNOTES] 1

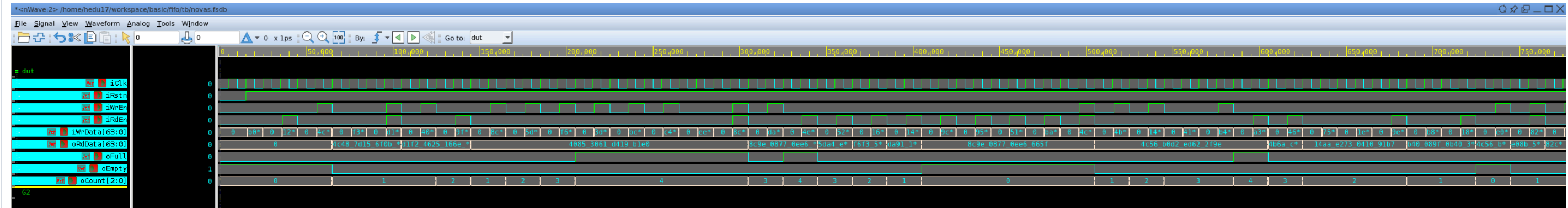
$finish called from file "/tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_root.svh", line 527.
$finish at simulation time 2515000

-----
VCS Coverage Metrics: during simulation line, cond, FSM, branch, tgl was monitored
-----

Coverage status: End of All Coverages ...

VCS Simulation Report
Time: 2515000 ps
CPU Time: 0.220 seconds; Data structure size: 0.6Mb
Mon May 11 15:14:47 2026
```


FIFO 결과



FIFO Verdi waveform

Summary

Hierarchy

Modules

Groups

Asserts

Statistics

Tests

Avg. Group Score:100.00% U+C:23 U:0 C:23 X:0

Avg. Group Inst. Score:100.00% U+C:23 U:0 C:23 X:0

Group	Score	Instances	U+C	U	C	X	Goal	Weight	AtLeast	PerInst	Overlap	AutoBinl	Missing	Comment
Cg tb_fifo_pkg::fifo_coverag...	100.00%	100.00%	23	0	23	0	100%	1	1	1	1	64	64	
Cp cp_cmd	100.00%		4	0	4	0	100%	1	1		1	0		
Cp cp_count	100.00%		3	0	3	0	100%	1	1		1	0		
Cp cp_empty	100.00%		2	0	2	0	100%	1	1		1	0		
Cp cp_full	100.00%		2	0	2	0	100%	1	1		1	0		
Ch cross_cmd_status	100.00%		12	0	12	0	100%	1	1				0	
Cg tb_fifo.me.obj.fifo_cg	100.00%		23	0	23	0	100%	1	1	1	0	64	64	

UVM result summary

```
UVM_INFO fifo_coverage.sv(50) @ 2515000: uvm_test_top.env.cov [FIFO_COV] Functional coverage = 100.00%
UVM_INFO /tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_report_server.svh(904) @ 2515000: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

** Report counts by severity
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UVM_FATAL : 0
** Report counts by id
[FIFO_COV] 1
[FIFO_SCB] 119
[RNTST] 1
[TEST DONE] 1
[UVM/RELNOTES] 1

$finish called from file "/tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_root.svh", line 527.
$finish at simulation time 2515000

-----
VCS Coverage Metrics: during simulation line, cond, FSM, branch, tgl was monitored
-----

Coverage status: End of All Coverages ...

VCS Simulation Report
Time: 2515000 ps
CPU Time: 0.220 seconds; Data structure size: 0.6Mb
Mon May 11 15:14:47 2026
```

Coverage result

블록별 검증 관점 비교

Block	RTL 특성	Monitor timing	Scoreboard model	Coverage 중심
Adder	조합 회로	입력 drive 후 #1 sample	expected = iA + iB	operand corner, carry
RAM	synchronous single-port memory	command edge sample, read data #1 sample	model_mem[DEPTH]	cmd, address, data
FIFO	pointer/count 기반 queue	edge 직전 status/data sample	model_q[\$]	cmd, full/empty, count

조합 회로

- 상태 모델 불필요
- 출력 폭과 carry 확인

Memory

- write/read 순서 모델링
- registered read timing 반영

Queue

- data order와 status 동시 검증
- blocked request 처리

결론

최종 정리

Adder, RAM, FIFO는 동일한 UVM testbench 골격을 공유하지만, 실제 검증 품질은 DUT별 timing rule, reference model, coverage 목표를 분리했을 때 확보된다.

구조 재사용

- test / env / agent / sequence 구조를 세 DUT에 동일하게 적용
- factory 기반 class 생성 흐름과 connect 구조 확인

모델링 차이

- Adder는 즉시 계산 reference model 사용
- RAM은 model_mem, FIFO는 model_q로 상태 유지

검증 증거

- Verdi waveform으로 pin-level timing 확인
- UVM report와 coverage 화면으로 scenario 수행 결과 확인

구분	이번 실습에서 확인한 점	추가 개선 방향
Sequence	directed case와 random case를 함께 구성	constraint 분리, corner regression 확대
Scoreboard	DUT 특성별 reference model을 분리	mismatch log format과 debug trace 정리
Coverage / SVA	기능 coverage와 protocol assertion을 함께 사용	covergroup 목표값과 assertion 항목 문서화